

Cloud Computing

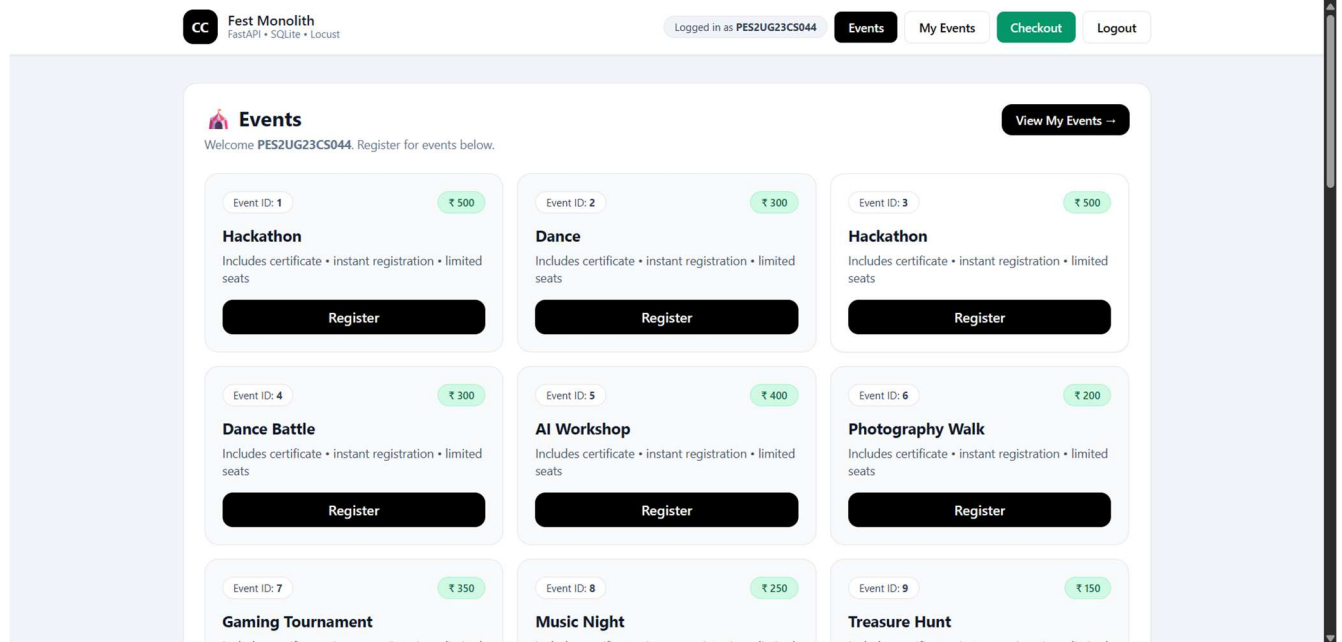
LAB 2- MONOLITHIC ARCHITECTURE

GitHub Link :

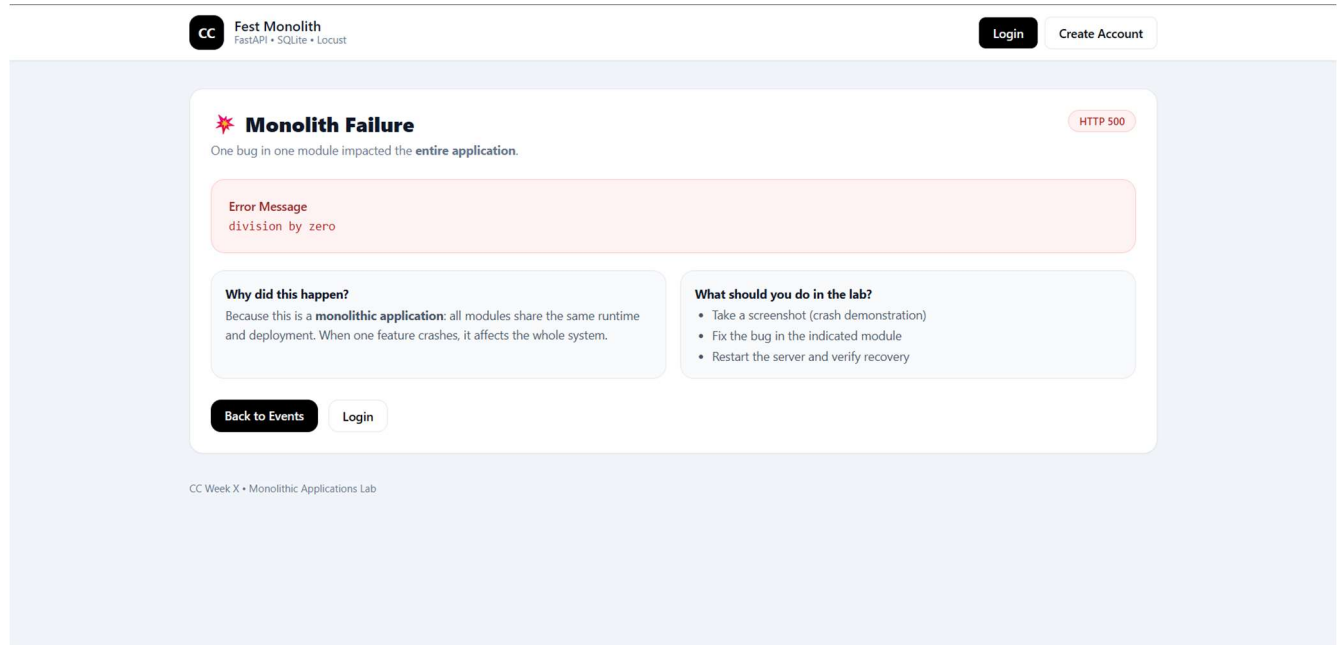
<https://github.com/akshayag2005/CLOUD-COMPUTING.git>



1. Screenshot-SS1 (Events page loaded)



2. Screenshot-SS2



```
INFO: 127.0.0.1:55482 - "GET /checkout HTTP/1.1" 500 Internal Server Error
ERROR: Exception in ASGI application
```

3. Screenshot-SS3

CC

Fest Monolith
FastAPI • SQLite • Locust

LoginCreate Account

Checkout

This route is used to demonstrate a monolith crash + optimization.

Total Payable
₹ 12400

After fixing + optimizing checkout logic, re-run Locust and compare results.

What you should observe

- One buggy feature can crash the entire monolith.
- Inefficient loops cause high response times under load.
- Optimization improves performance but architecture still scales as one unit.

Next Lab: Split this monolith into Microservices (Events / Registration / Checkout).

CC Week X • Monolithic Applications Lab

INFO: 127.0.0.1:62818 - "GET /checkout HTTP/1.1" 200 OK

4. Screenshot-SS4

LOCUST

Host
http://localhost:8000

Status
CLEANUP

RPS
0.7

Failures
0%

EDIT

STOP

RESET

STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	19	0	8	2100	2100	115.94	6	2068	2798	0.7	0
	Aggregated	19	0	8	2100	2100	115.94	6	2068	2798	0.7	0

[2026-01-20 23:04:49,164] AKSHAY/INFO/locust.main: Starting locust 2.43.1

[2026-01-20 23:04:49,164] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.

[2026-01-20 23:05:51,300] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second

[2026-01-20 23:05:51,304] AKSHAY/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)

[2026-01-20 23:14:32,984] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second

[2026-01-20 23:14:32,986] AKSHAY/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)

Traceback (most recent call last):

File "D:\SEM 6\Cloud computing\CC Lab\PES2UG23CS044\venv\Lib\site-packages\gevent\ffi\loop.py", line 279, in python_check_callback

def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt

2026-01-20T17:45:26Z

[2026-01-20 23:15:26,724] AKSHAY/INFO/locust.main: Shutting down (exit code 0)

Type	Name	# reqs	# fails	Avg	Min	Max	Med	req/s	failures/s
GET	/checkout	19	0(0.00%)	115	4	2041	8	0.66	0.00
	Aggregated	19	0(0.00%)	115	4	2041	8	0.66	0.00

Response time percentiles (approximated)

Type	Name	50%	66%	75%	80%	90%	95%	98%	99%	99.9%	99.99%	100%	# reqs
GET	/checkout	8	10	10	11	18	2000	2000	2000	2000	2000	2000	19
	Aggregated	8	10	10	11	18	2000	2000	2000	2000	2000	2000	19

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5. Screenshot-SS5

LOCUST

Host: http://localhost:8000 | Status: CLEANUP | RPS: 0.8 | Failures: 0% | EDIT | STOP | RESET | ⚙️

STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	19	0	9	2100	2100	119.07	4	2074	2798	0.8	0
Aggregated		19	0	9	2100	2100	119.07	4	2074	2798	0.8	0

Traceback (most recent call last):
File "D:\SEM 6\Cloud computing\CC Lab\PES2UG23CS044\venv\Lib\site-packages\gevent_ffi\loop.py", line 279, in python_check_callback
def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T17:53:42Z
[2026-01-20 23:23:42,862] AKSHAY/INFO/locust.main: Shutting down (exit code 0)

Type	Name	# reqs	# fails	Avg	Min	Max	Med	req/s	failures/s
GET	/checkout	19	0(0.00%)	119	4	2073	9	0.66	0.00
Aggregated		19	0(0.00%)	119	4	2073	9	0.66	0.00

Response time percentiles (approximated)

Type	Name	50%	66%	75%	80%	90%	95%	98%	99%	99.9%	99.99%	100%	# reqs
GET	/checkout	9	9	20	21	23	2100	2100	2100	2100	2100	2100	19
Aggregated		9	9	20	21	23	2100	2100	2100	2100	2100	2100	19

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6. Screenshot BEFORE optimization-SS6

LOCUST

Host: http://localhost:8000 | Status: STOPPED | RPS: 0.5 | Failures: 0% | NEW | RESET | ⚙️

STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	15	0	360	2500	2500	481.08	216	2483	37438	0.5	0
Aggregated		15	0	360	2500	2500	481.08	216	2483	37438	0.5	0

Traceback (most recent call last):
File "D:\SEM 6\Cloud computing\CC Lab\PES2UG23CS044\venv\Lib\site-packages\gevent_ffi\loop.py", line 279, in python_check_callback
def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument

KeyboardInterrupt
2026-01-20T17:57:40Z
[2026-01-20 23:27:40,847] AKSHAY/INFO/locust.main: Shutting down (exit code 0)

Type	Name	# reqs	# fails	Avg	Min	Max	Med	req/s	failures/s
GET	/events?user=locust_user	15	0(0.00%)	481	215	2483	360	0.52	0.00
Aggregated		15	0(0.00%)	481	215	2483	360	0.52	0.00

Response time percentiles (approximated)

Type	Name	50%	66%	75%	80%	90%	95%	98%	99%	99.9%	99.99%	100%	# reqs
GET	/events?user=locust_user	360	360	390	410	430	2500	2500	2500	2500	2500	2500	15
Aggregated		360	360	390	410	430	2500	2500	2500	2500	2500	2500	15

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7. Screenshot AFTER optimization-SS7

Host: http://localhost:8000 | Status: CLEANUP | RPS: 0.7 | Failures: 0% | EDIT | STOP | RESET

STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	18	0	8	2100	2100	124.76	5	2079	37438	0.7	0
Aggregated		18	0	8	2100	2100	124.76	5	2079	37438	0.7	0

```
[2026-01-20 23:36:37,572] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:36:37,572] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:36:50,408] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:36:50,412] AKSHAY/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloud computing\CC Lab\PES2UG23CS044\venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument
KeyboardInterrupt
[2026-01-20 23:37:57,668] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type      Name                                     # reqs  # fails | Avg  Min  Max  Med | req/s  failures/s
-----
GET       /events?user=locust_user                18      0(0.00%) | 124   4  2079   8 | 0.63    0.00
Aggregated                                     18      0(0.00%) | 124   4  2079   8 | 0.63    0.00

Response time percentiles (approximated)
Type      Name                                     50%    66%    75%    80%    90%    95%    98%    99%    99.9%  99.99%  100% #
-----
GET       /events?user=locust_user                8      9     10    14    28    2100  2100  2100  2100  2100  18
Aggregated                                     8      9     10    14    28    2100  2100  2100  2100  2100  18
```

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8. Screenshot BEFORE optimization-SS8

Host: http://localhost:8000 | Status: CLEANUP | RPS: 0.6 | Failures: 0% | EDIT | STOP | RESET

STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	17	0	110	2200	2200	249.62	90	2236	3144	0.6	0
Aggregated		17	0	110	2200	2200	249.62	90	2236	3144	0.6	0

```
[2026-01-20 23:41:21,105] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:41:21,105] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:41:36,529] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:41:36,531] AKSHAY/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloud computing\CC Lab\PES2UG23CS044\venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument
KeyboardInterrupt
[2026-01-20 23:42:34,055] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type      Name                                     # reqs  # fails | Avg  Min  Max  Med | req/s  failures/s
-----
GET       /my-events?user=locust_user             17      0(0.00%) | 249   90  2235  110 | 0.57    0.00
Aggregated                                     17      0(0.00%) | 249   90  2235  110 | 0.57    0.00

Response time percentiles (approximated)
Type      Name                                     50%    66%    75%    80%    90%    95%    98%    99%    99.9%  99.99%  100% # reqs
-----
GET       /my-events?user=locust_user             110    130   130   150   240   2200  2200  2200  2200  2200  17
Aggregated                                     110    130   130   150   240   2200  2200  2200  2200  2200  17
```

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9. Screenshot AFTER optimization-SS9



Host

http://localhost:8000

Status

CLEANUP

RPS

0.5

Failures

0%

EDIT

STOP

RESET



STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	18	0	7	2100	2100	121.07	4	2054	3144	0.5	0
Aggregated		18	0	7	2100	2100	121.07	4	2054	3144	0.5	0

```
>
[2026-01-20 23:45:41,533] AKSHAY/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 23:45:41,533] AKSHAY/INFO/locust.main: Starting web interface at http://localhost:8089, press enter to open your default browser.
[2026-01-20 23:45:59,236] AKSHAY/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 23:45:59,237] AKSHAY/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)
Traceback (most recent call last):
  File "D:\SEM 6\Cloud computing\CC Lab\PES2UG23CS044\.venv\Lib\site-packages\gevent\_ffi\loop.py", line 279, in python_check_callback
    def python_check_callback(self, watcher_ptr): # pylint:disable=unused-argument
KeyboardInterrupt
2026-01-20T18:16:57Z
[2026-01-20 23:46:57,072] AKSHAY/INFO/locust.main: Shutting down (exit code 0)
Type      Name                                     # reqs   # fails | Avg   Min   Max   Med | req/s  failures/s
-----
GET       /my-events?user=locust_user              18      0(0.00%) | 121   3    2053  7 | 0.61   0.00
-----
Aggregated                                     18      0(0.00%) | 121   3    2053  7 | 0.61   0.00
-----

Response time percentiles (approximated)
Type      Name                                     50%    66%    75%    80%    90%    95%    98%    99%    99.9%  99.99%  100% # reqs
-----
GET       /my-events?user=locust_user                8      9      9     10     12    2100    2100    2100    2100    2100    2100  18
-----
Aggregated                8      9      9     10     12    2100    2100    2100    2100    2100    2100  18
-----

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```

• Route 1: /events

1. What was the bottleneck?

The /events route contained inefficient logic where event data was processed repeatedly and unnecessary looping was performed while constructing the response. This increased CPU usage for every request.

2. What change did you make?

The code was optimized by removing redundant loops and processing the event data in a single pass. Only the required fields were accessed directly instead of repeatedly recalculating values.

3. Why did the performance improve?

By reducing unnecessary computations and iterations, the CPU workload per request decreased. This resulted in lower response time and improved overall performance during load testing.

• Route 2: /my-events

1. What was the bottleneck?

The /my-events route performed repeated filtering and processing of the same dataset for every request, leading to inefficient use of CPU resources.

2. What change did you make?

The logic was optimized by simplifying the filtering process and avoiding repeated scans of the data. The required information was computed more efficiently in a single operation.

3. Why did the performance improve?

Optimizing the filtering logic reduced redundant processing, which lowered execution time per request. As a result, response time improved when tested using Locust.